

# Math Check #2 — Loops by Hand

OPIM 5641 · Business Decision Modeling  
Studio 2 warm-up · 5 points · ~8 min

Name: \_\_\_\_\_ NetID: \_\_\_\_\_

By hand — no laptop. Loops are the engine of everything we do this week: if you can trace one on paper, you can debug it on screen. Work it out with a pencil, take a photo of your finished page, and upload it to HuskyCT.

## Q1 • What does this print? 2 pts

```
1 total = 0
2 for i in range(3):
3     total = total + 10
4     print(total)
```

## Q2 • Now a loop inside a loop 2 pts

Write every line it prints, *in order*:

```
1 for size in ['small', 'large']:
2     for color in ['red', 'blue']:
3         print(size, color)
```

## Q3 • Count without writing them out 1 pt

How many lines does this print? Show the arithmetic — do *not* list them.

```
1 for chairs in range(3):
2     for desks in range(4):
3         for tables in range(2):
4             print(chairs, desks, tables)
```